
Student Performance Prediction System

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1. Problem Description

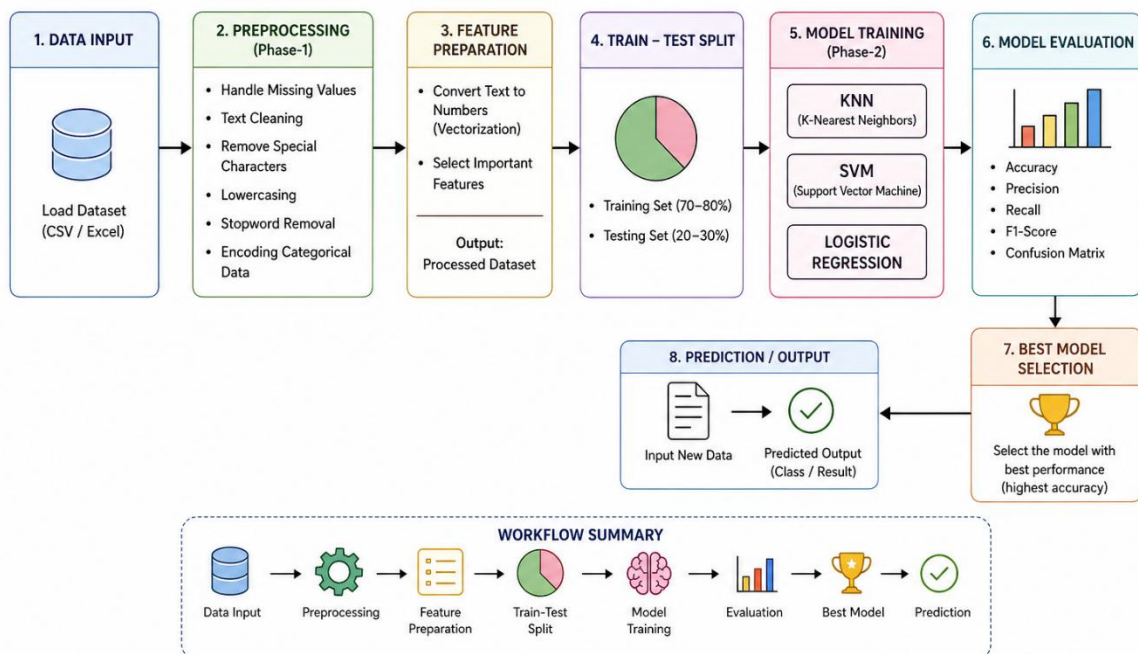
With the rapid growth of online platforms, users frequently share their opinions and reviews about products and services. These reviews contain valuable information that can help businesses understand customer satisfaction and improve their offerings. However, manually analyzing large volumes of text data is time-consuming and inefficient.

There is a need for an automated system that can accurately classify user reviews into different sentiment categories such as positive, negative, and neutral. This helps in quick decision-making and enhances user experience.

The problem addressed in this project is to develop a machine learning-based sentiment analysis model that processes textual review data, cleans and transforms it into numerical form, and classifies it using algorithms such as Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). The system aims to achieve reliable performance in predicting sentiment based on user reviews and ratings.

2. Architecture

ARCHITECTURE OF THE PROGRAM



3. Models Used(Applied/ Planned) Give a brief introduction of the models used

1. K-Nearest Neighbors (KNN)

Introduction:

KNN is a simple and easy-to-understand machine learning algorithm used for classification and prediction.

How it works:

- It finds the 'K' nearest data points to a new input.
- The output is decided based on the majority class of those neighbors.

Why used in this project:

- Works well with structured data
- Simple and effective for basic classification tasks

2. Support Vector Machine (SVM)

Introduction:

SVM is a powerful supervised learning algorithm used for classification and regression tasks.

How it works:

- It finds a boundary (hyperplane) that separates different classes.
- Tries to maximize the margin between classes for better accuracy.

Why used in this project:

- High accuracy in many cases
- Effective for both linear and non-linear data

3. Logistic Regression

Introduction:

Logistic Regression is a statistical model used for binary classification problems.

How it works:

- It predicts the probability of an output class (0 or 1).
- Uses a sigmoid function to convert values into probabilities.

Why used in this project:

- Fast and efficient
- Easy to implement and interpret
- Works well for classification tasks

4. Evaluation metrics used

1. Accuracy

Introduction:

Accuracy measures how many predictions made by the model are correct.

Formula:

$$\text{Accuracy} = (\text{Correct Predictions}) / (\text{Total Predictions})$$

Why used:

- Gives overall performance of the model
- Easy to understand

2. Precision

Introduction:

Precision measures how many of the predicted positive values are actually correct.

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Why used:

- Important when false positives should be minimized

3. Recall

Introduction:

Recall measures how many actual positive cases are correctly identified.

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Why used:

- Important when missing positive cases is risky

4. F1-Score

Introduction:

F1-Score is the harmonic mean of Precision and Recall.

Formula:

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Why used:

- Balances both precision and recall
- Useful when data is imbalanced

5. Confusion Matrix

Introduction:

A confusion matrix is a table used to evaluate model performance.

It shows:

- TP (True Positive)
- TN (True Negative)
- FP (False Positive)
- FN (False Negative)

Why used:

- Gives a complete picture of prediction results

5. Preliminary Results and Visualizations

Logistic Regression

Accuracy: 0.8629032258064516

	precision	recall	f1-score	support
Negative	1.00	1.00	1.00	50
Neutral	0.00	0.00	0.00	34
Positive	0.83	1.00	0.91	164
accuracy			0.86	248
macro avg	0.61	0.67	0.64	248
weighted avg	0.75	0.86	0.80	248

KNN

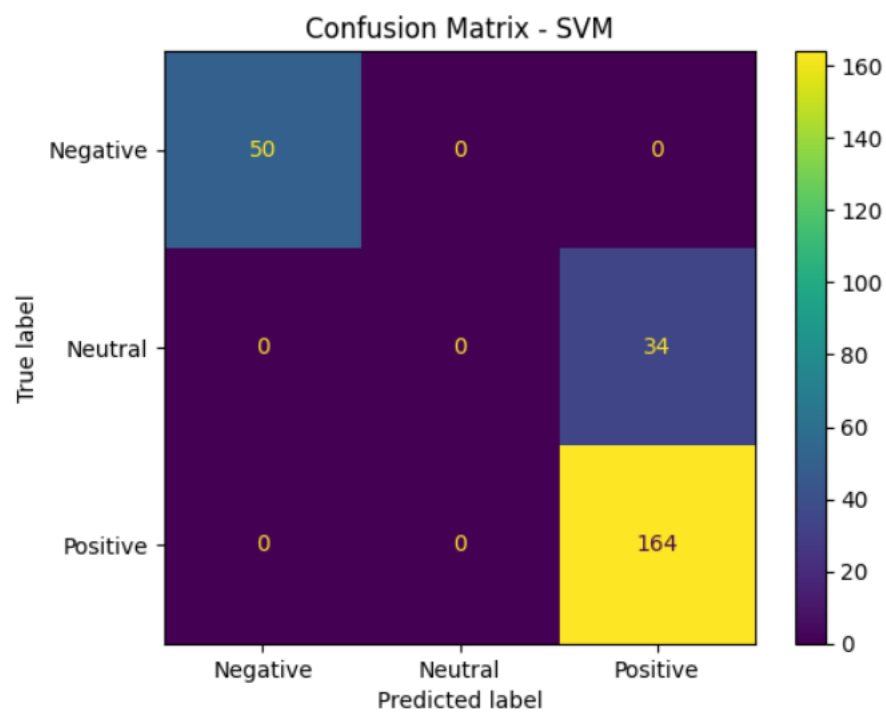
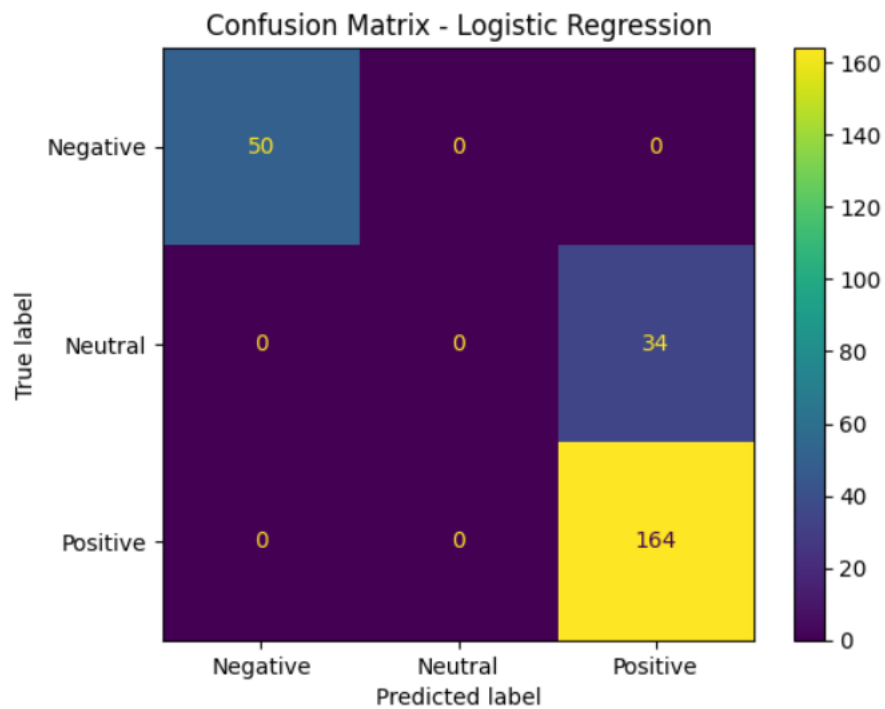
Accuracy: 0.9758064516129032

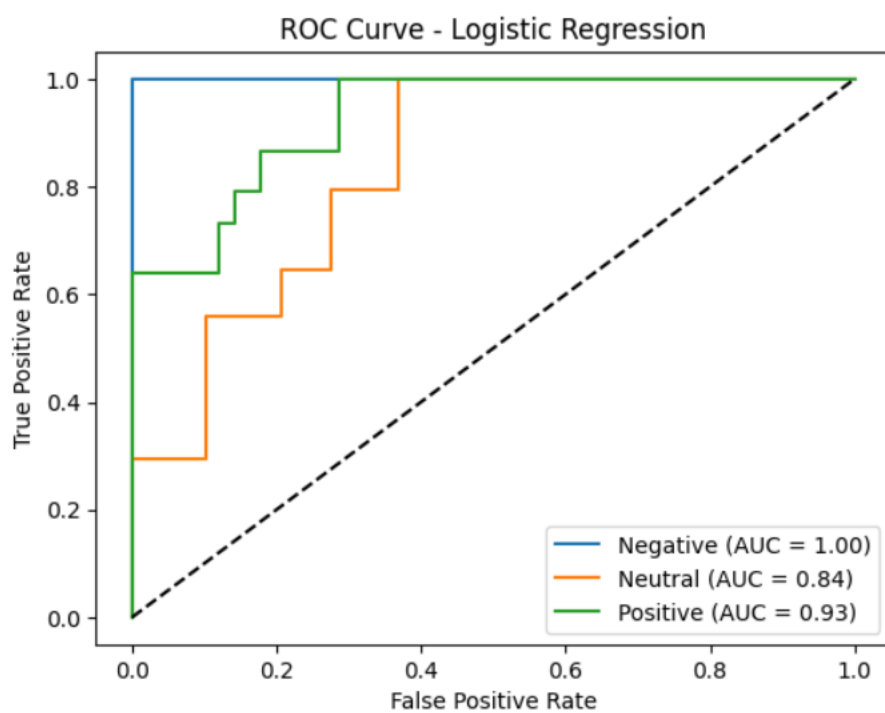
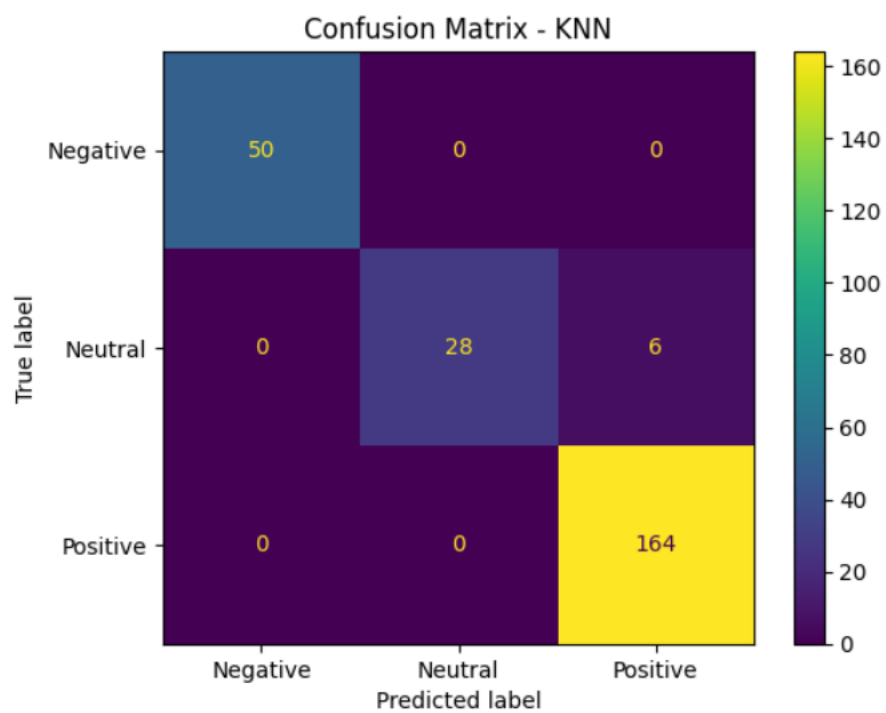
	precision	recall	f1-score	support
Negative	1.00	1.00	1.00	50
Neutral	1.00	0.82	0.90	34
Positive	0.96	1.00	0.98	164
accuracy			0.98	248
macro avg	0.99	0.94	0.96	248
weighted avg	0.98	0.98	0.97	248

SVM

Accuracy: 0.8629032258064516

	precision	recall	f1-score	support
Negative	1.00	1.00	1.00	50
Neutral	0.00	0.00	0.00	34
Positive	0.83	1.00	0.91	164
accuracy			0.86	248
macro avg	0.61	0.67	0.64	248
weighted avg	0.75	0.86	0.80	248





6. References:

Books:

1. Let Us Python – Yashavant Kanetkar

Online Resources:

- GeeksforGeeks – <https://www.geeksforgeeks.org/>
- TutorialsPoint – <https://www.tutorialspoint.com/cprogramming/index.htm>
- Stack Overflow – <https://stackoverflow.com>